

Mathematical and Computational Methods

Exercise Sheet 4: Vectors

This sheet accompanies *Unit 4 (Vectors)*. All methods are those of Unit 4 only—vector algebra, the dot and cross products, and the equations of lines and planes; nothing from later units is needed.

How to use this sheet. Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showssolutions{}\input{MCM_Exercises_Unit4}"`) and recompile.

Notation. Vectors are bold, $\mathbf{u}, \mathbf{v}, \mathbf{w}$; $\|\mathbf{v}\|$ is the magnitude, $\hat{\mathbf{v}} = \mathbf{v} / \|\mathbf{v}\|$ a unit vector, $\mathbf{u} \cdot \mathbf{v}$ the dot product, $\mathbf{u} \times \mathbf{v}$ the cross product, and $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}$ the projection of $\mathbf{u}$ onto $\mathbf{v}$. The standard basis of $\mathbb{R}^3$ is $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

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1 Warm-up drills

Short, routine problems — at least one per skill — to build fluency. Answers should be quick.

Exercise 1.1 (*★ drill— component arithmetic*). Let $\mathbf{u} = (3, -1, 2)$ and $\mathbf{v} = (-1, 4, 0)$. Compute

$$(a) \mathbf{u} + \mathbf{v}, \quad (b) \mathbf{u} - \mathbf{v}, \quad (c) 3\mathbf{u}, \quad (d) 2\mathbf{u} - 3\mathbf{v}.$$

Exercise 1.2 (*★ drill— vector between points, distance*). For the points $A(2, -1, 3)$ and $B(5, 1, -1)$, find (a) the vector $\overrightarrow{AB}$ and (b) the distance from A to B .

Exercise 1.3 (*★ drill— unit vectors*). Normalise each vector: (a) $\mathbf{v} = (3, 4)$, (b) $\mathbf{w} = (1, -2, 2)$.

Exercise 1.4 (*★ drill— dot product and orthogonality*). Let $\mathbf{a} = (2, 3)$, $\mathbf{b} = (3, -2)$, $\mathbf{c} = (1, 1)$. Compute $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$ and $\mathbf{b} \cdot \mathbf{c}$, and say which pair (if any) is orthogonal.

Exercise 1.5 (*★ drill— angle via the dot product*). Find the angle between (a) $(3, 0)$ and $(0, -5)$; (b) $(1, 1)$ and $(-1, 0)$.

Exercise 1.6 (*★ drill— projection*). Project $\mathbf{u} = (6, 2)$ (a) onto $\mathbf{v} = (3, 0)$ and (b) onto $\mathbf{w} = (0, 5)$.

Exercise 1.7 (*★ drill— cross product*). Compute (a) $\mathbf{i} \times \mathbf{j}$ and (b) $(1, 2, 3) \times (4, 5, 6)$.

Exercise 1.8 (*★ drill— area*). Find the area of the parallelogram spanned by $\mathbf{u} = (2, -2, 1)$ and $\mathbf{v} = (1, 2, 2)$, and hence the area of the triangle with these as two sides.

Exercise 1.9 (*★ drill— equation of a line*). (a) Give the vector, parametric and symmetric equations of the line through $P(1, 2, 3)$ in the direction $\mathbf{d} = (2, -1, 4)$. (b) Give the vector equation and the slope of the line through $A(0, 1)$ and $B(4, 3)$ in $\mathbb{R}^2$.

Exercise 1.10 (*★ drill— equation of a plane*). (a) Find the Cartesian equation of the plane through $(2, -1, 3)$ with normal $\mathbf{n} = (1, 2, -1)$. (b) Write down a normal vector to the plane $3x - y + 2z = 7$.

Exercise 1.11 (*★ drill, conceptual — true or false*). State whether each is true or false, with a one-line reason.

1. If $\mathbf{u} \cdot \mathbf{v} = 0$ then $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$.
2. $\mathbf{u} \times \mathbf{v} = \mathbf{v} \times \mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$.
3. $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ for every $\mathbf{v} \in \mathbb{R}^3$.
4. The dot product of two vectors is a vector.
5. $\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$ for all $\mathbf{u}, \mathbf{v}$.

Exercise 1.12 (*★ drill— sketching sums and differences*). Take $\mathbf{u} = (3, 1)$ and $\mathbf{v} = (1, 2)$ in $\mathbb{R}^2$. On a single set of axes, sketch $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$, each drawn from the origin. Read the two combinations off your sketch and confirm them by component arithmetic.

2 Tutorial problems

The core set for the 2-hour tutorial: standard difficulty, anchored on the unit's worked examples, with problems aimed at each cohort (mathematics, data science, physics). Problems flagged ► form the live *tutorial core* to be worked in the 2-hour class; the remaining tutorial problems are for completion at home.

Exercise 2.1 (► **core—linear combinations). Let $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, -1)$.

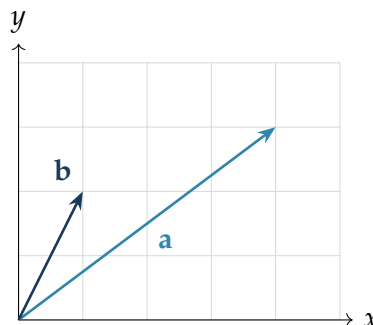
1. Express $\mathbf{c} = (7, 0)$ as $\alpha\mathbf{a} + \beta\mathbf{b}$.
2. Are $\mathbf{a}$ and $\mathbf{b}$ linearly independent? Explain geometrically.

Exercise 2.2 (► **core, data science — angle and cosine similarity). Over the four-word vocabulary (data, model, train, plot) two short documents have word-count vectors $\mathbf{u} = (1, 1, 0, 0)$ and $\mathbf{v} = (0, 1, 1, 0)$.

1. Find the angle θ between $\mathbf{u}$ and $\mathbf{v}$ using the dot product.
2. State their cosine similarity $\cos\theta$ and interpret it: would these documents be judged similar?

Exercise 2.3 (► **core—parallel and perpendicular components). Resolve $\mathbf{u} = (2, 4)$ into a component parallel to $\mathbf{v} = (1, 1)$ and a component perpendicular to it. Verify the two pieces are orthogonal and sum to $\mathbf{u}$.

Exercise 2.4 (► **core—reading a diagram). The figure shows two vectors $\mathbf{a}$ and $\mathbf{b}$ drawn from the origin on a unit grid.



1. Read off the components of $\mathbf{a}$ and $\mathbf{b}$ from the grid.
2. Compute the projection $\text{proj}_{\mathbf{b}} \mathbf{a}$ of $\mathbf{a}$ onto $\mathbf{b}$.

Exercise 2.5 (► **core—a perpendicular direction). Find a vector orthogonal to both $\mathbf{a} = (1, 2, 1)$ and $\mathbf{b} = (2, -1, 3)$, and then a *unit* vector in that direction.

Exercise 2.6 (**core—area of a triangle in space). Find the area of the triangle with vertices $A(1, 1, 0)$, $B(2, 3, 1)$ and $C(0, 2, 2)$.

Exercise 2.7 (► **core—a line, and points on it). Let ℓ be the line through $A(1, 0, 2)$ and $B(3, 4, 6)$.

1. Give a parametric equation of ℓ , using the simplest integer direction.
2. Does $C(0, -2, 0)$ lie on ℓ ?
3. Does $D(2, 1, 4)$ lie on ℓ ?

Exercise 2.8 (► **core— *the plane through three points (anchor)*). Find the equation of the plane through $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 6)$, in both point-normal and Cartesian form, and verify all three points satisfy it.

Exercise 2.9 (**core— *the same plane in parametric form*). Using the *same* three points $A(2, 0, 0)$, $B(0, 3, 0)$, $C(0, 0, 6)$ as above:

1. Write the plane in parametric (vector) form $\mathbf{r} = A + s \overrightarrow{AB} + t \overrightarrow{AC}$.
2. Evaluate the point at $s = 1$, $t = 1$ and verify it satisfies the Cartesian equation $3x + 2y + z = 6$ found above.

Exercise 2.10 (**core— *line meets plane*). Where does the line $\mathbf{r} = t(1, 1, 1)$ meet the plane $3x + 2y + z = 6$ found above?

Exercise 2.11 (► **core, physics — *torque and work*). A force $\mathbf{F} = (2, 3, 1)$ (newtons) acts at the point with position vector $\mathbf{r} = (1, 0, 2)$ (metres) measured from a pivot.

1. Find the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ about the pivot.
2. The point of application then undergoes the displacement $\mathbf{d} = (0, 4, 0)$ (metres). Find the work $W = \mathbf{F} \cdot \mathbf{d}$ done by the force.

Exercise 2.12 (**core, data science — *ranking by cosine similarity*). A query is the vector $\mathbf{q} = (1, 1, 1)$. Three documents are $\mathbf{d}_1 = (2, 2, 2)$, $\mathbf{d}_2 = (1, 1, 0)$, $\mathbf{d}_3 = (0, 0, 5)$. Rank them by cosine similarity to $\mathbf{q}$, and comment on $\mathbf{d}_1$.

Exercise 2.13 (**core, enrichment — *volume and coplanarity*). 1. Find the volume of the parallelepiped on $\mathbf{u} = (1, 0, 1)$, $\mathbf{v} = (2, 3, 0)$, $\mathbf{w} = (1, 1, 1)$ using the scalar triple product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$.

2. Show that $\mathbf{a} = (1, 1, 1)$, $\mathbf{b} = (2, 1, 0)$, $\mathbf{c} = (0, 1, 2)$ are coplanar.

Exercise 2.14 (**core, conceptual — *spot the error*). A student writes: “For $\mathbf{u} = (1, 2, 2)$ and $\mathbf{v} = (2, 2, 1)$ we have $\mathbf{u} \cdot \mathbf{v} = 2 + 4 + 2 = 8$, so $\cos \theta = 8$ and $\theta = \arccos 8$.” Identify the mistake and give the correct angle.

3 Further practice (home)

A larger bank for independent study, from routine to harder, ending with a few **challenge** problems (***). Levels are tagged on each.

Exercise 3.1 (*drill—parallel vectors). Find the scalar k for which $(k, 6)$ is parallel to $(2, 3)$.

Exercise 3.2 (*drill—magnitude in higher dimensions). For $\mathbf{v} = (1, 1, 1, 1) \in \mathbb{R}^4$ find $\|\mathbf{v}\|$ and the unit vector $\hat{\mathbf{v}}$.

Exercise 3.3 (*drill—perimeter from distances). Find the perimeter of the triangle with vertices $A(0, 0)$, $B(3, 0)$, $C(0, 4)$.

Exercise 3.4 (**core—unit vectors at a given angle). Find all unit vectors in $\mathbb{R}^2$ that make an angle of 45° with $(1, 0)$.

Exercise 3.5 (**core—lengths from dot products). Given $\|\mathbf{u}\| = 3$, $\|\mathbf{v}\| = 5$ and $\mathbf{u} \cdot \mathbf{v} = 6$, find $\|\mathbf{u} + \mathbf{v}\|$ and $\|\mathbf{u} - \mathbf{v}\|$.

Exercise 3.6 (**core—angle in a triangle). For the triangle $A(1, 0)$, $B(3, 2)$, $C(0, 1)$, find the interior angle at vertex A .

Exercise 3.7 (**core—solve for orthogonality). Find k so that $(k, 2, 1)$ and $(3, k, -5)$ are orthogonal.

Exercise 3.8 (**core—resolve into components). Resolve $\mathbf{u} = (3, 4)$ into parts parallel and perpendicular to $\mathbf{v} = (2, 1)$.

Exercise 3.9 (**core—anticommutativity and area). For $\mathbf{u} = (2, 1, -1)$, $\mathbf{v} = (1, 3, 2)$, compute $\mathbf{u} \times \mathbf{v}$ and $\mathbf{v} \times \mathbf{u}$, confirm they are negatives, and give the area of the triangle on $\mathbf{u}, \mathbf{v}$.

Exercise 3.10 (**core—unit normal to a plane). Find a unit normal to the plane through $P(1, 0, 0)$, $Q(0, 1, 0)$, $R(0, 0, 1)$, and the Cartesian equation of the plane.

Exercise 3.11 (**core—intersection of two lines). Do the lines $\ell_1 : \mathbf{r} = t(1, 2)$ and $\ell_2 : \mathbf{r} = (3, 0) + s(-1, 1)$ meet? If so, where?

Exercise 3.12 (**core—parallel planes). Are the planes $2x - y + 3z = 5$ and $-4x + 2y - 6z = 1$ parallel? Do they intersect?

Exercise 3.13 (**core—plane parallel to a given one). Find the plane through $(1, 2, 3)$ parallel to $2x - y + 2z = 0$.

Exercise 3.14 (**core—angle between two planes). Find the acute angle between the planes $x + y + z = 1$ and $x + z = 0$.

Exercise 3.15 (**core—distance from a point to a plane). Find the distance from the point $P(3, 1, 2)$ to the plane $2x - 2y + z = 4$, and the foot F of the perpendicular from P to the plane.

Exercise 3.16 (**core, physics—equilibrium of forces). Two forces $\mathbf{F}_1 = (2, 3)$ and $\mathbf{F}_2 = (-1, 4)$ (newtons) act on a particle. Find the third force $\mathbf{F}_3$ that holds it in equilibrium, and its magnitude.

Exercise 3.17 (**core, data science—orthogonal preferences). Two users rate four films, giving rating-deviation vectors $\mathbf{a} = (2, 2, 2, 2)$ and $\mathbf{b} = (1, -1, 1, -1)$. Compute their cosine similarity and interpret it.

Exercise 3.18 (**core, physics — work done sliding down a ramp). A crate of mass 10 kg slides down a ramp, undergoing the displacement $\mathbf{s} = (6, -8)$ (metres), with the $+y$ axis pointing up. Taking $g = 10 \text{ m/s}^2$, the weight is the force $\mathbf{F} = (0, -100)$ (newtons).

1. Find the work $W = \mathbf{F} \cdot \mathbf{s}$ done by gravity.
2. Find the component of the weight in the direction of motion.

Exercise 3.19 (**core— a “show that” bridge to the proofs). These identities power the two proofs that follow; establish them first. Using only $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w}$ and the bilinearity of the dot product, show that for all $\mathbf{u}, \mathbf{v}$

$$(a) \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2, \quad (b) \|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2\mathbf{u} \cdot \mathbf{v} + \|\mathbf{v}\|^2.$$

Exercise 3.20 (***challenge— the parallelogram law). Using only the properties of the dot product, prove that for all $\mathbf{u}, \mathbf{v}$

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2.$$

Exercise 3.21 (***challenge— diagonals of a rhombus). A rhombus has sides $\mathbf{u}$ and $\mathbf{v}$ with $\|\mathbf{u}\| = \|\mathbf{v}\|$. Its diagonals are $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$. Prove the diagonals are perpendicular.

Exercise 3.22 (***challenge— coplanarity of four points). Decide whether the points $P(0,0,0)$, $Q(1,1,0)$, $R(1,0,1)$, $S(0,1,1)$ are coplanar. If not, give the volume of the parallelepiped on $\overrightarrow{PQ}, \overrightarrow{PR}, \overrightarrow{PS}$.

Exercise 3.23 (***challenge— foot of the perpendicular). Let ℓ be the line through the origin in the direction $\mathbf{d} = (1,2)$, and let $P(5,0)$. Find the foot F of the perpendicular from P to ℓ , and the distance from P to ℓ .

Coverage map (for tutors)

Each syllabus Skill and worked-example anchor for Unit 4, with the exercises that exercise it.

Skill / anchor	Exercises
Vector operations & geometric interpretation	1.1, 1.2, 1.11, 1.12, 2.1, 2.4, 3.1, 3.5, 3.16
Magnitudes & unit vectors	1.2, 1.3, 3.2, 3.3, 3.4
Dot product (angle, projection, orthogonality)	1.4, 1.5, 1.6, 2.2, 2.3, 2.4, 2.12, 2.14, 3.5–3.8, 3.17, 3.19, 3.20, 3.21, 3.23
Cross product (normal, area)	1.7, 1.8, 2.5, 2.6, 3.9, 3.10
Lines (vector / parametric / symmetric)	1.9, 2.7, 3.11
Planes (normal / Cartesian / parametric, distance)	1.10, 2.8, 2.9, 2.10, 3.10, 3.12, 3.13, 3.14, 3.15
<i>Anchor</i> : angle between two vectors via the dot product	1.5, 2.2, 3.6, (2.14)
<i>Anchor</i> : the plane through three points	2.8, 2.9, 3.10
Enrichment: scalar triple product (volume / coplanarity)	2.13, 3.22
$\mathbb{R}^n$ / cosine similarity (data science)	2.2, 2.12, 3.2, 3.17
Physics applications (work, torque, equilibrium)	2.11, 3.16, 3.18
Conceptual / true–false / spot-the-error	1.11, 2.14

Quick answer key
